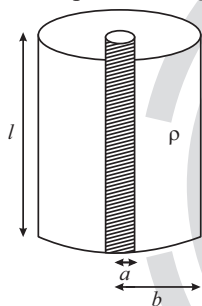


Current Electricity

Single Correct Type Questions

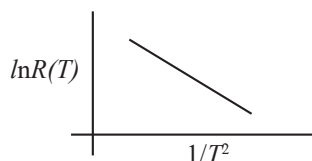
1. Model a torch battery of length l to be made up of a thin cylindrical bar of radius ' a ' and a concentric thin cylindrical shell of radius ' b ' filled in between with an electrolyte of resistivity ρ (see figure). If the battery is connected to a resistance of value R , the maximum Joule heating in R will take place for [3 Sep, 2020 (Shift-I)]



- (a) $R = \frac{\rho}{\pi l} \ln\left(\frac{b}{a}\right)$ (b) $R = \frac{2\rho}{\pi l} \ln\left(\frac{b}{a}\right)$
 (c) $R = \frac{\rho}{2\pi l} \ln\left(\frac{b}{a}\right)$ (d) $R = \frac{\rho}{2\pi l} \ln\left(\frac{a}{b}\right)$

2. In an experiment, the resistance of a material is plotted as a function of temperature (in some range). As shown in the figure, it is a straight line. One may conclude that:

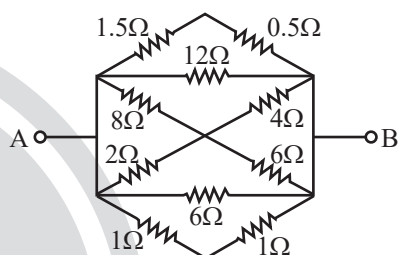
[10 April, 2019 (Shift-I)]



- (a) $R(T) = R_0 e^{T^2/T_0^2}$ (b) $R(T) = \frac{R_0}{T^2}$
 (c) $R(T) = R_0 e^{-T^2/T_0^2}$ (d) $R(T) = R_0 e^{-(T_0^2/T^2)}$

3. The equivalent resistance between A and B is _____

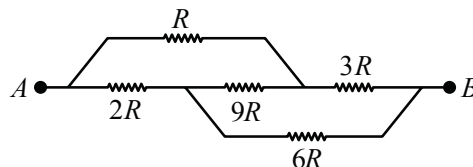
[30 Jan, 2023 (Shift-II)]



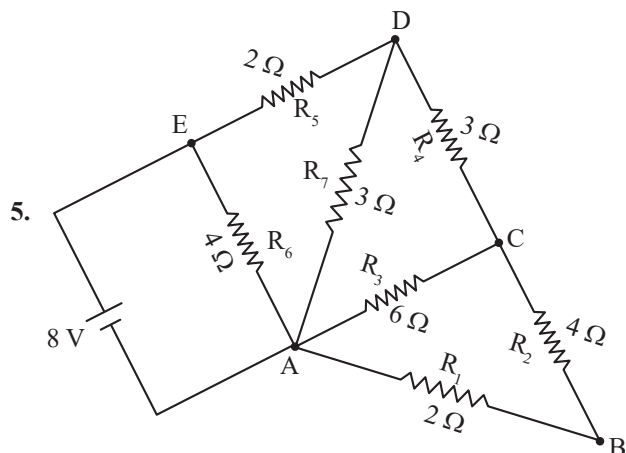
- (a) $\frac{2}{3} \Omega$
 (b) $\frac{1}{2} \Omega$
 (c) $\frac{3}{2} \Omega$
 (d) $\frac{1}{3} \Omega$

4. The equivalent resistance between A and B of the network shown in figure:

[01 Feb, 2023 (Shift-I)]



- (a) $11\frac{2R}{3}$
 (b) $14 R$
 (c) $21 R$
 (d) $\frac{8}{3} R$



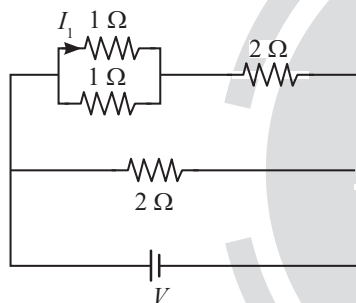
The current flowing through R_2 is:

[11 April, 2023 (Shift-II)]

- (a) $\frac{2}{3} A$ (b) $\frac{1}{4} A$ (c) $\frac{1}{2} A$ (d) $\frac{1}{3} A$

6. The current I_1 (in A) flowing through 1Ω resistor in the following circuit is

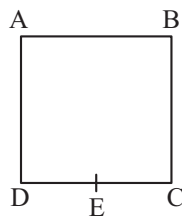
[7 Jan, 2020 (Shift-I)]



- (a) 0.25 (b) 0.4 (c) 0.2 (d) 0.5

7. A wire of resistance R is bent to form a square ABCD as shown in the figure. The effective resistance between E and C is: (E is mid-point of arm CD)

[9 April, 2019 (Shift-I)]



- (a) $\frac{1}{16} R$ (b) $\frac{7}{64} R$ (c) $\frac{3}{4} R$ (d) R

8. An electric bulb of 500 watt at 100 volt is used in a circuit having a 200 V supply. Calculate the resistance R to be connected in series with the bulb so that the power delivered by the bulb is 500 W. [26 Aug, 2021 (Shift-II)]

- (a) 20Ω (b) 5Ω (c) 10Ω (d) 30Ω

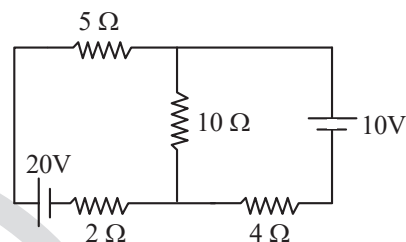
9. Two equal resistances when connected in series to a battery, consume electric power of $60 W$. If these resistances are now connected in parallel combination to the same battery, the electric power consumed will be.

[11 Jan, 2019 (Shift-I)]

- (a) $60 W$
(b) $240 W$
(c) $120 W$
(d) $30 W$

10. In the figure shown, the current in the $10 V$ battery is close to:

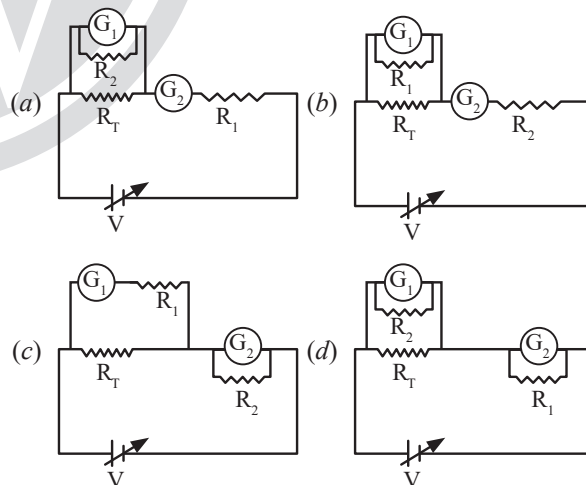
[6 Sep, 2020 (Shift-II)]



- (a) 0.71 A from positive to negative terminal
(b) 0.42 A from positive to negative terminal
(c) 0.21 A from positive to negative terminal.
(d) 0.36 A from negative to positive terminal.

11. A student is provided with a variable voltage source V , a test resistor $R_t = 10 \Omega$, two identical galvanometers G_1 and G_2 and two additional resistors, $R_1 = 10 M \Omega$ and $R_2 = 0.001 \Omega$. For conducting an experiment to verify ohms law, the most suitable circuit is:

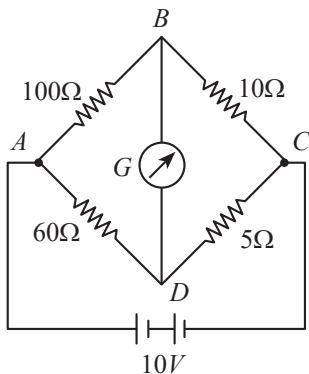
[6 April, 2023 (Shift-II)]



12. The four arms of a Wheatstone bridge have resistances as shown in the figure. A galvanometer of 15Ω resistance is connected across BD. Calculate the current through the galvanometer when a potential difference of $10 V$ is maintained across AC.

[17 March, 2021 (Shift-II)]

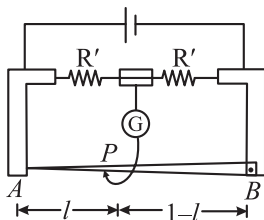
- (a) $4.87 \mu A$ (b) $2.44 mA$
(c) $4.87 mA$ (d) $2.44 \mu A$



13. A moving coil galvanometer, having a resistance G , produces full scale deflection when a current I_g flows through it. This galvanometer can be converted into (i) an ammeter of range 0 to I_0 ($I_0 > I_g$) by connecting a shunt resistance R_A to it and (ii) into a voltmeter of range 0 to V ($V = GI_0$) by connecting a series resistance R_V to it. Then,

[12 April, 2019 (Shift-II)]

- (a) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \left(\frac{I_0 - I_g}{I_g} \right)^2$
 (b) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g} \right)^2$
 (c) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \frac{I_g}{I_0 - I_g}$
 (d) $R_A R_V = G^2$ and $\frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g} \right)^2$
14. In a meter bridge, the wire of length 1 m has a non-uniform cross-section such that the variation $\frac{dR}{dl}$ of its resistance R with length l is $\frac{dR}{dl} \propto \frac{1}{\sqrt{l}}$. Two equal resistances are connected as shown in the figure. The galvanometer has zero deflection when the jockey is at point P . What is the length AP ? [12 Jan, 2019 (Shift-I)]



- (a) 0.2 m
 (b) 0.3 m
 (c) 0.25 m
 (d) 0.35 m

15. In a Wheatstone bridge (see figure) Resistance P and Q are approximately equal. When $R = 400 \Omega$, the bridge is balanced. On interchanging P and Q , the value of R , for balance is 405Ω . The value of X is close to:

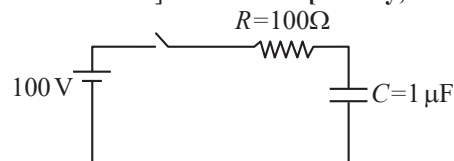
[11 Jan, 2019 (Shift-I)]

- (a) 401.5 ohm (b) 404.5 ohm
 (c) 403.5 ohm (d) 402.5 ohm

16. A capacitor of capacitance $C = 1 \mu F$ is suddenly connected to a battery of 100 volt through a resistance $R = 100 \Omega$. The time taken for the capacitor to be charged to get 50 V is:

[Take $\ln 2 = 0.69$]

[27 July, 2021 (Shift-I)]



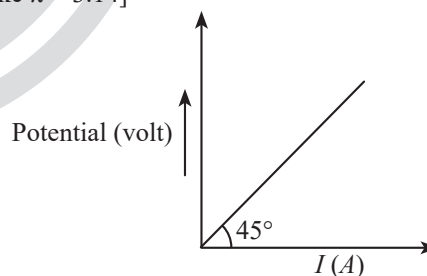
- (a) 3.33×10^{-4} s
 (b) 1.44×10^{-4} s
 (c) 0.30×10^{-4} s
 (d) 0.69×10^{-4} s

Integer Type Questions

17. The variation of applied potential and current flowing through a given wire is shown in figure. The length of wire is 31.4 cm. The diameter of wire is measured as 2.4 cm. The resistivity of the given wire is measured as $x \times 10^{-3} \Omega \text{cm}$. The value of x is _____.

[29 June, 2022 (Shift-I)]

[Take $\pi = 3.14$]

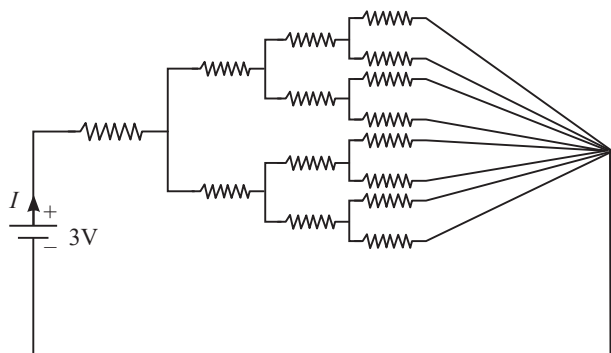


18. The current density in a cylindrical wire of radius 4 mm is $4 \times 10^6 \text{ Am}^{-2}$. The current through the outer portion of the wire between radial distance $\frac{R}{2}$ and R is _____ πA .

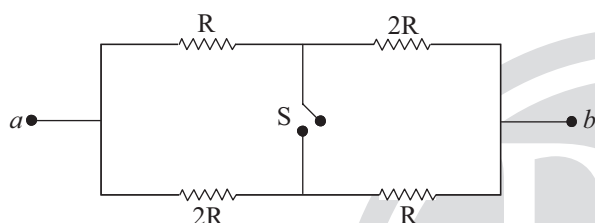
[27 June, 2022 (Shift-I)]

19. All resistances in figure are 1Ω each. The value of current 'I' is $\frac{a}{5} A$. The value of a is _____.

[28 June, 2022 (Shift-II)]

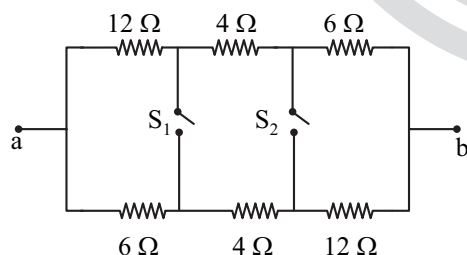


20. The ratio of the equivalent resistance of the network (shown in figure) between the points a and b when switch is open and switch is closed is $x : 8$. The value of x is _____.

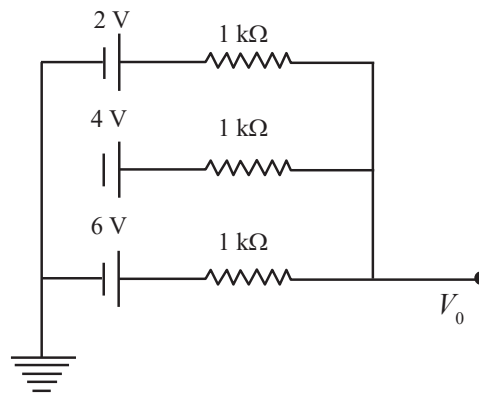


21. First, a set of n equal resistors of 10Ω each are connected in series to a battery of emf 20V and internal resistance 10Ω . A current I is observed to flow. Then, the n resistors are connected in parallel to the same battery. It is observed that the current is increased 20 times, then the value of n is _____.

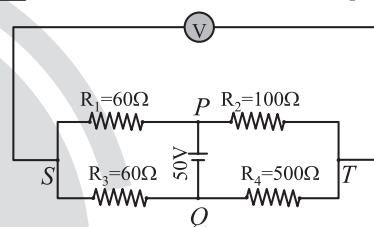
22. In the given figure switches S_1 and S_2 are in open condition. The resistance across ab when the switches S_1 and S_2 are closed is _____ Ω .



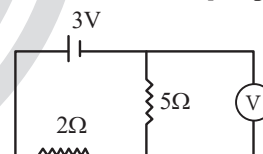
23. A resistor dissipates 192J of energy in 1s when a current of 4A is passed through it. Now, when the current is doubled, the amount of thermal energy dissipated in 5s is _____ J .
24. In the given figure, the value of V_0 will be _____ V .



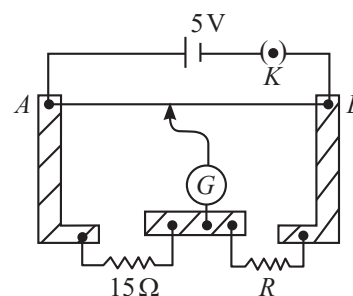
25. In the balanced condition, the values of the resistances of the four arms of a Wheatstone bridge are shown in the figure below. The resistance R_3 has temperature coefficient $0.0004\text{ }^\circ\text{C}^{-1}$. If the temperature of R_3 is increased by $100\text{ }^\circ\text{C}$, the voltage developed between S and T will be _____ volt.



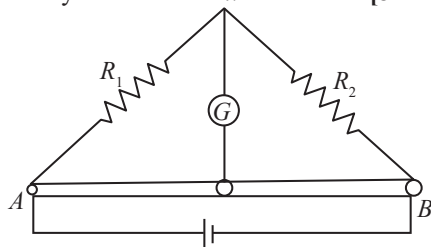
26. As shown in the figure, the voltmeter reads 2V across 5Ω resistor. The resistance of the voltmeter is _____ Ω .



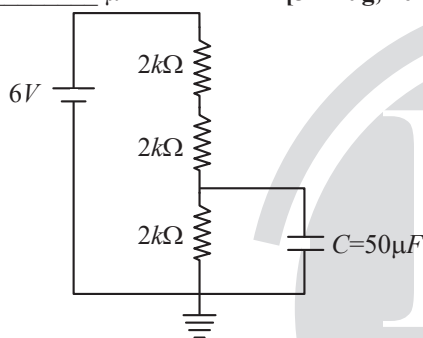
27. A meter bridge setup is shown in the figure. It is used to determine an unknown resistance R using a given resistor of 15Ω . The galvanometer (G) shows null deflection when tapping key is at 43 cm mark from end A . If the end correction for end A is 2 cm . Then the determined value of R will be _____ Ω .



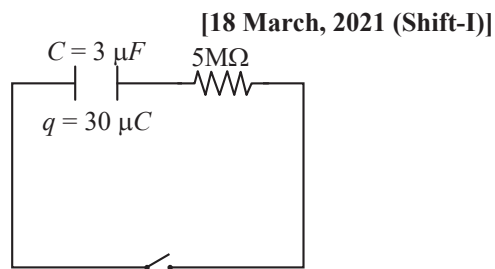
28. Two resistances $R_1 = X \Omega$ and $R_2 = 1 \Omega$ are connected to a wire AB of uniform resistivity, as shown in the figure. The radius of the wire varies linearly along its axis from 0.2 mm at A to 1 mm at B . A galvanometer (G) connected to the center of the wire, 50 cm from each end along its axis, shows zero deflection when A and B are connected to a battery. The value of x is [JEE Adv, 2022]



29. A capacitor of $50 \mu\text{F}$ is connected in a circuit as shown in figure. The charge on the upper plate of the capacitor is _____ μC . [31 Aug, 2021 (Shift-I)]



30. The circuit shown in the figure consists of a charged capacitor of capacity $3\mu\text{F}$ and a charge of $30 \mu\text{C}$. At time $t = 0$, when the key is closed, the value of current flowing through the $5M\Omega$ resistor is ' x ' μA . [18 March, 2021 (Shift-I)]



The value of ' x ' to the nearest integer is _____.

ANSWER KEY

- | | | | | | | | | | |
|----------|----------|------------|---------|--------------------|----------|-----------|----------|-----------|---------|
| 1. (c) | 2. (d) | 3. (a) | 4. (d) | 5. (d) | 6. (c) | 7. (b) | 8. (a) | 9. (b) | 10. (c) |
| 11. (c) | 12. (c) | 13. (b) | 14. (c) | 15. (d) | 16. (d) | 17. [144] | 18. [48] | 19. [8] | 20. [9] |
| 21. [20] | 22. [10] | 23. [3840] | 24. [4] | 25. [0.26 to 0.28] | 26. [20] | 27. [19] | 28. [5] | 29. [100] | |
| 30. [2] | | | | | | | | | |

EXPLANATIONS

1. (c) Power in external $R = \left(\frac{\varepsilon}{r+R} \right)^2 R$

This power is maximum, when $R = r$

$$r = \int_a^b \rho \frac{dx}{2\pi x l}$$

$$r = \frac{\rho}{2\pi l} \ln \frac{b}{a}$$

2. (d) Here, we have given resistance as a function of temperature, $R(T)$

$$\frac{1}{T^2} + \frac{\ln R(T)}{\ln R(T_0)} = 1$$

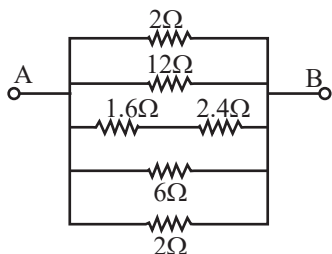
Now,

$$\Rightarrow \ln R(T) = \ln R(T_0) \left(1 - \frac{T_0^2}{T^2} \right)$$

$$R(T) = R(T_0) e^{-\left(\frac{T_0}{T}\right)^2}$$

Hence, option (d) is correct answer.

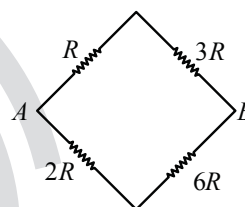
3. (a) Equivalent circuit can be drawn as



$$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{2} + \frac{1}{12} + \frac{1}{4} + \frac{1}{6} + \frac{1}{2} = \frac{3}{2}$$

$$\Rightarrow R_{eq} = \frac{2}{3} \Omega$$

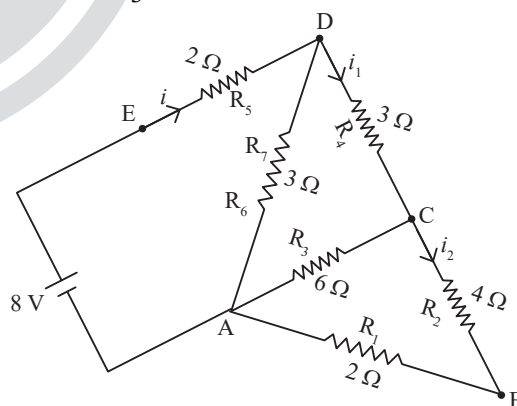
4. (d) As the wheatstone bridge is in balanced condition, no current will flow through the $9R$.



$$\Rightarrow \frac{1}{R_{eq}} = \frac{1}{4R} + \frac{1}{8R}$$

$$\Rightarrow R_{eq} = \frac{8R}{3}$$

5. (d)



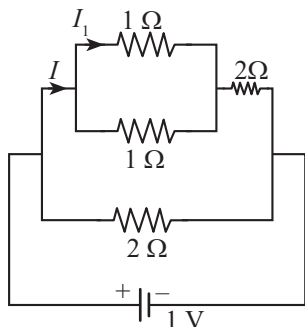
$$R_{eq} = 4\Omega$$

$$i = \frac{8}{4} = 2A \quad i_1 = \frac{2 \times 3}{3+6} = \frac{2}{3}A$$

$$i_2 = \frac{2/3}{2} = \frac{1}{3}A$$

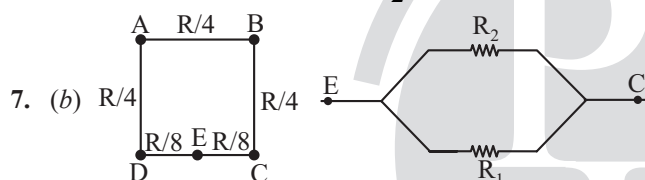
6. (c) Resistance in parallel $= \frac{R_1 R_2}{R_1 + R_2} = \frac{1}{2} \Omega$

Now, resistance in series $R' = \frac{1}{2} + 2 = 2.5 \Omega$



Current through upper branch $I = \frac{1}{2.5} = 0.4 A$

Current through 1Ω (I_1) $= \frac{I}{2} = 0.2 A$



$$R_1 = \frac{R}{8}, R_2 = \frac{R}{8} + \frac{R}{4} + \frac{R}{4} + \frac{R}{4} + \frac{R}{4} = \frac{7R}{8}$$

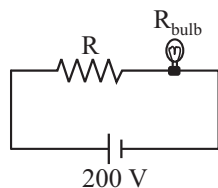
$$\frac{1}{R_{EC}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{1}{R_{EC}} = \frac{8}{7R} + \frac{8}{R}$$

$$R_{EC} = \frac{7R}{8+56} = \frac{7R}{64}$$

8. (a) $R_{\text{bulb}} = \frac{V^2}{P_{\text{bulb}}} = \frac{100 \times 100}{500} = 20 \Omega$

$$I = \frac{V_{\text{supply}}}{R_{\text{bulb}} + R} = \frac{200}{20 + R}$$



According to the question.

Power delivered by bulb $= 500 W$

$$\Rightarrow I^2 R_{\text{bulb}} = 500$$

$$\Rightarrow \left(\frac{200}{20 + R} \right)^2 \times 20 = 500$$

$$\Rightarrow (R + 20)^2 = \frac{200 \times 200 \times 20}{500} = 1600$$

$$\Rightarrow R + 20 = 40$$

$$\therefore R = 20 \Omega$$

9. (b) When resistances are connected in parallel.

$$P_T = \frac{P_1 P_2}{P_1 + P_2} = 60$$

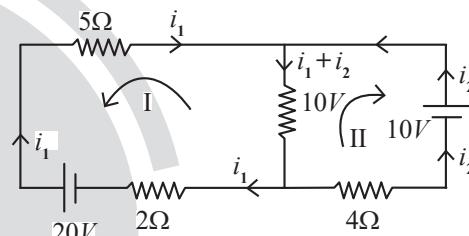
$$\therefore P_1 = P_2$$

$$\therefore P_1 = P_2 = 120 W$$

So total consumed power

$$P' = P_1 + P_2 = 120 + 120 = 240 W$$

10. (c)



Loop-I:

$$-20 + 2i_1 + 10(i_1 + i_2) + 5i_1 = 0$$

$$17i_1 + 10i_2 = 20$$

...(i)

Loop-II:

$$-10 + 4i_2 + 10(i_1 + i_2) = 0$$

$$10i_1 + 14i_2 = 10$$

...(ii)

$$\text{Equation (i)} \times (10) - \text{Equation (ii)} \times 17$$

$$170i_1 + 100i_2 = 200$$

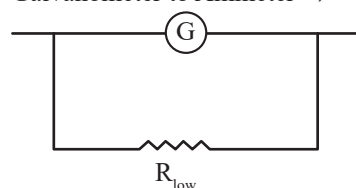
$$-170i_1 + 238i_2 = 170$$

$$-138i_2 = 30$$

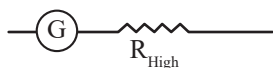
$$i_2 = -\frac{30}{138} = -0.217 A$$

"-ve" sign indicates that current flows from "+ve" to "-ve" terminal in 10 V battery.

11. (c) Galvanometer to Ammeter $\Rightarrow$



Galvanometer to Voltmeter $\Rightarrow$



12. (c) Let, $V_A = 10\text{ V}$, $V_B = x\text{ V}$, $V_C = 0\text{ V}$, and $V_D = y\text{ V}$
Net current from B is zero, so,

$$\frac{x-10}{100} + \frac{x-y}{15} + \frac{x-0}{10} = 0 \Rightarrow 53x - 20y = 30 \quad \dots (i)$$

Net current from D is zero, so,

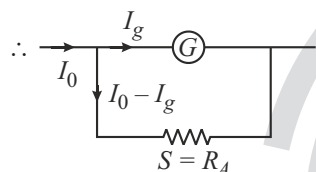
$$\frac{y-10}{60} + \frac{y-x}{15} + \frac{y-0}{5} = 0 \Rightarrow 17y - 4x = 10 \quad \dots (ii)$$

Solving equations (i) and (ii), we get.

$$x = 0.865 \text{ and } y = 0.792$$

$$i_G = \frac{x-y}{15} = 4.87\text{ mA}$$

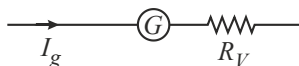
13. (b) When galvanometer is used as an ammeter, shunt is used in parallel with galvanometer.



$$\therefore I_g G = (I_0 - I_g) R_A$$

$$\therefore R_A = \left(\frac{I_g}{I_0 - I_g} \right) G$$

When galvanometer is used as a voltmeter, resistance is used in series with galvanometer.

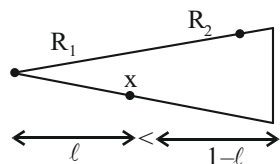


$$I_g(G + R_V) = V = GI_0 \text{ (given } V = GI_0)$$

$$\therefore R_V = \frac{(I_0 - I_g)G}{I_g}$$

$$\therefore R_A R_V = G^2 \text{ and } \frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g} \right)^2$$

14. (c) $\frac{R'}{R_1} = \frac{R'}{R_2} \Rightarrow R_1 = R_2 = \frac{R}{2}$



Total resistance = R

$$= \int_{\text{wire}} dR = \int_0^1 \frac{k}{\sqrt{l}} \cdot d\ell = k \left[2\sqrt{\ell} \right]_0^1$$

$$= 2k$$

$$R_1 = \frac{R}{2} = k \Rightarrow \int_0^1 \frac{k}{\sqrt{\ell}} \cdot d\ell = k$$

$$[2k\sqrt{\ell}]_0^1 = k$$

$$\sqrt{\ell} = \frac{1}{2}, \ell = 0.25\text{ m}$$

15. (d) We know that, $\frac{P}{Q} = \frac{R}{S}$

$$\text{Case 1: } \frac{P}{Q} = \frac{400}{x}$$

$$\text{Case 2: } \frac{Q}{P} = \frac{405}{x}$$

Using case 1 and case 2:

$$x^2 = 400 \times 405$$

$$x = 402.5\text{ }\Omega$$

16. (d) Given, $C = 1\text{ }\mu\text{F} = 1 \times 10^{-6}\text{ F}$

$$V_0 = 100\text{ volt}, V = 50\text{ volt}$$

$$R = 100\text{ }\Omega$$

$$\text{Where } \tau = RC = 100 \times 10^{-6}\text{ sec}$$

$$50V = 100V \left(1 - e^{-\frac{t}{10^{-4}}} \right)$$

$$\frac{1}{2} = 1 - e^{-10^4 t} \Rightarrow \frac{1}{2} = e^{-10^4 t}$$

$$2 = e^{+10^4 t} \Rightarrow \ln 2 = 10^4 t$$

$$t = \frac{\ln 2}{10^4}$$

$$t = 0.693 \times 10^{-4}\text{ sec.}$$

17. [144]

$$\text{We know that, } R = \frac{V}{I} = 1\Omega$$

$$R = \frac{\rho l}{A} \Rightarrow 1 = \frac{x \times 10^{-3} \times 31.4}{\pi \times (1.2)^2}$$

$$\Rightarrow x = 144$$

18. [48] $i = A \times j$

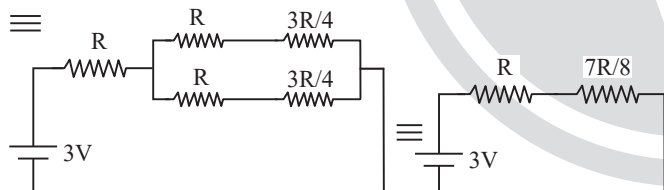
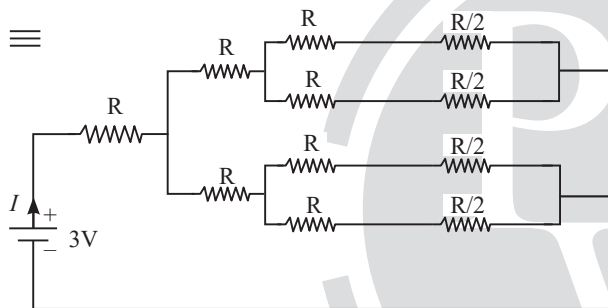
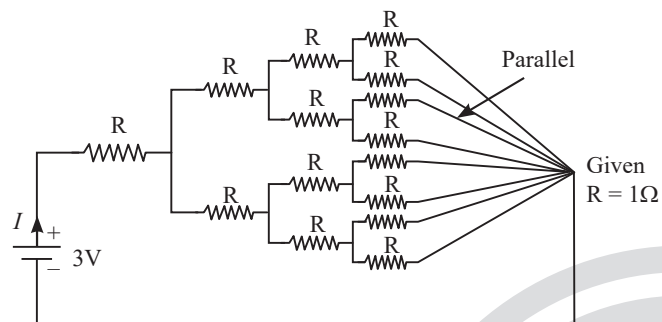
$$= \pi \left(R^2 - \frac{R^2}{4} \right) j$$

$$= \frac{3\pi R^2}{4} \times j$$

$$= \frac{3\pi \times (4 \times 10^{-3})^2}{4} \times 4 \times 10^6$$

$$= 48\pi$$

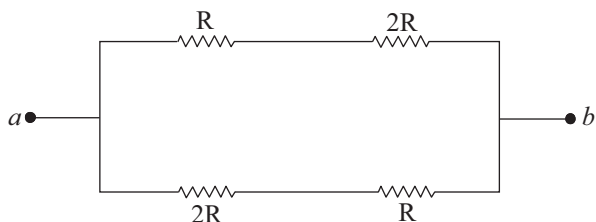
19. [8]



$$\text{Current, } I = \frac{3 \times 8}{15R} = \frac{8}{5R}$$

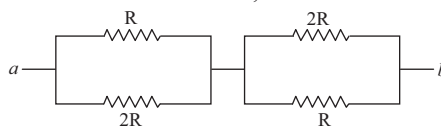
On comparing above result with $I = a/5$, we get
 $a = 8$

20. [9] When switch is open,



$$R_{ab} = \frac{3R}{2}$$

When switch is closed,



$$R'_{ab} = 2 \times \frac{R \times 2R}{R + 2R} = \frac{4R}{3}$$

$$\frac{R_{ab}}{R'_{ab}} = \frac{3R/2}{4R/3} = \frac{9}{8} = \frac{x}{8} \Rightarrow x = 9$$

21. [20] In series connection, $R_{eq} = nR = 10n$

$$\Rightarrow I_{se} = \frac{20}{10 + 10n} = \frac{2}{1 + n}$$

In parallel connection, $R_{eq} = \frac{10}{n}$

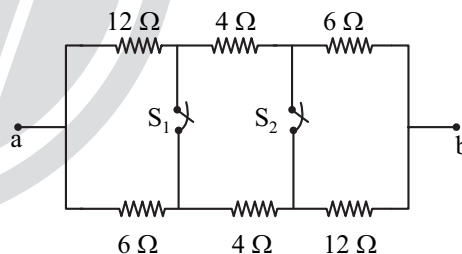
$$\Rightarrow I_p = \frac{20}{\frac{10}{n} + 10} = \frac{2n}{1 + n}$$

According to the question,

$$\Rightarrow \frac{I_p}{I_{se}} = 20 = \frac{2n/1+n}{2/1+n}$$

$$\Rightarrow n = 20$$

22. [10] The equivalent resistance,



$$R_{eq} = (12 \parallel 6) + (4 \parallel 4) + (4 \parallel 12)$$

$$= \left(\frac{12 \times 6}{18} \right) + \left(\frac{4 \times 4}{8} \right) + \left(\frac{6 \times 12}{18} \right) = 4 + 2 + 4 = 10\Omega$$

23. [3840] $E = i_2 R t$

$$\Rightarrow 192 = (4)^2 R (1)$$

$$\Rightarrow R = 12\Omega$$

As current is doubled and time = 5s

So, energy dissipated in this case $E' = 3840 \text{ J}$

24. [4] From Nodal analysis,

$$\frac{V_0 - 2}{1k\Omega} + \frac{V_0 - 4}{1k\Omega} + \frac{V_0 - 6}{1k\Omega} = 0$$

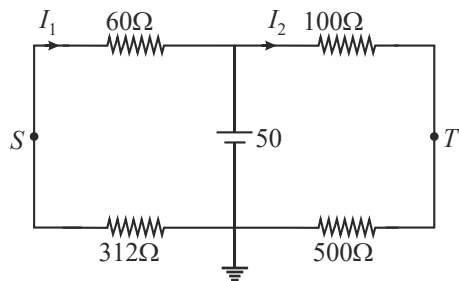
$$V_0 = 4 \text{ V}$$

25. [0.26 to 0.28]

$$R_3' = 300(1 + \alpha \Delta T)$$

$$= 312 \Omega$$

Now

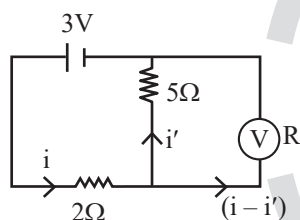


$$I_1 = \frac{50}{372} \text{ and } I_2 = \frac{50}{600}$$

$$V_S - V_T = 312 I_1 - 500 I_2$$

$$= 41.94 - 41.67 = -0.27 \text{ V}$$

26. [20] $i' = \frac{2V}{5\Omega} = \frac{2}{5} \text{ A}$



$$i = \frac{(3-2)}{2} = \frac{1}{2} \text{ A}$$

$$\therefore \text{Current through voltmeter} = \frac{1}{2} - \frac{2}{5} = \frac{1}{10} \text{ A}$$

$$\therefore \text{For voltmeter } V = (i - i')R$$

$$2 = \left(\frac{1}{10}\right)R \Rightarrow R = 20\Omega$$

27. [19] Unknown resistance R

Known resistance, $P = 15\Omega$

At null point

$$\frac{P}{R} = \frac{l}{100 - l}$$

$$\frac{15}{R} = \frac{43 + 2}{100 - (43)}$$

$$R = 19\Omega$$

28. [5]

$$R_{AO} = \int_0^{1/2} \frac{\rho}{\pi (r + 4rx)^2} dx = \frac{2\rho}{3\pi r^2}$$

$$R_{OB} = \int_{1/2}^1 \frac{\rho}{\pi (r + 4rx)^2} dx = \frac{2\rho}{15\pi r^2}$$

$$(x) \frac{2\rho}{15\pi r^2} = (1) \frac{2\rho}{3\pi r^2} \Rightarrow x = 5$$

29. [100] We know that, charge on the capacitor, $Q = CV$

$$Q = 50 \mu\text{F} \times 2\text{V} = 100 \times 10^{-6} \text{ C}$$

$$Q = 100 \mu\text{C}$$

Hence, charge on the upper plate of capacitor is $100 \mu\text{C}$

30. [2] $I = \frac{V}{R} = \frac{30/3}{5 \times 10^6} = 2 \times 10^{-6}$